

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied to improve the operation of signalized intersections by responding to changing traffic conditions. Conventional traffic signals may operate using fixed timing plans, while adaptive signal-control systems use traffic measurements to adjust signal timing. Recent research has also investigated machine vision and artificial intelligence (AI) for estimating traffic conditions and supporting adaptive signal control. This paper presents a conceptual architecture for a local-AI traffic signal system in which camera data are processed near the intersection, reducing the need to transmit complete video streams to a remote server. The proposed architecture also considers emergency-vehicle signal preemption as an additional component. Unlike the unsupported claim that emergency-vehicle path clearing is generally unavailable, established traffic-signal systems already provide emergency-vehicle preemption. Therefore, the research opportunity is better framed as integrating reliable local computer-vision detection with existing signal-preemption mechanisms and coordinating the response across multiple intersections.

Index Terms—smart traffic lights, artificial intelligence, edge computing, computer vision, adaptive signal control, emergency vehicle preemption

I. INTRODUCTION

Traffic congestion is an important problem in urban transportation systems. Traffic signal control is one of the mechanisms used to manage vehicle movements at intersections. Conventional signal systems can use fixed timing plans or coordinated timing schedules. These approaches are relatively predictable, but their performance can be affected when traffic demand differs substantially from the conditions for which the timing plans were developed [3].

Adaptive signal control provides an alternative by using traffic measurements to modify signal timing according to current traffic conditions. The Federal Highway Administration describes adaptive signal control as a technology that captures current traffic-demand information and uses it to adjust signal timing. Such systems can use different sensing, communication, and processing configurations depending on the application [3], [4].

Artificial intelligence and machine learning have also been investigated for traffic signal control. In particular, deep reinforcement learning has been studied as an approach to traffic signal control, with research covering different learning architectures, simulation environments, and design considerations [1]. Computer vision provides another possible source of traffic information. Gorodokin *et al.* investigated machine-

vision-based vehicle detection and traffic-flow parameters for adaptive traffic-light control [2].

This paper focuses on a local-AI architecture in which traffic-camera information is processed close to the intersection. The objective is not to claim that local processing is always faster or cheaper than cloud processing, but to investigate whether local processing can reduce communication requirements and support timely traffic-control decisions.

II. BACKGROUND AND RELATED WORK

A. Adaptive Traffic Signal Control

Traditional traffic-signal operation can include fixed-cycle coordination and other predetermined timing strategies. Fixed timing plans may work adequately when traffic conditions are relatively predictable, but changing traffic demand can reduce their effectiveness [4].

Adaptive signal control systems collect traffic information and use it to update signal timing. Depending on the implementation, optimization can be performed at different time scales and may require different levels of traffic detection, communication, and computing resources [3].

Research has also examined learning-based traffic signal control. Rasheed *et al.* reviewed the application of deep reinforcement learning to traffic signal control and discussed its potential, simulation platforms, complexity, and open research problems [1].

B. Machine Vision for Traffic Monitoring

Camera-based systems can provide information about vehicles and traffic conditions without requiring every measurement to be obtained from conventional road sensors. Gorodokin *et al.* investigated machine vision using YOLOv4 to track traffic-flow parameters and proposed an adaptive traffic-light-control methodology based on observed traffic characteristics [2].

These studies indicate that computer vision can be used as an input to adaptive traffic-signal control. However, the actual performance of a practical system depends on camera placement, lighting, weather, detection accuracy, processing hardware, communication infrastructure, and the signal-controller interface.

III. LOCAL AI AND EDGE PROCESSING

Cloud-based architectures can transmit camera information to a remote server for processing. Such an architecture can

provide access to substantial computational resources, but it also requires communication between the roadside system and the remote computing infrastructure.

A local or edge-computing architecture instead performs at least part of the processing near the traffic intersection. A local computer can receive camera data, estimate traffic conditions, and send a control request or traffic measurement to the signal-control system. Processing locally may reduce the amount of raw video that must be transmitted over a network. It can also reduce dependence on continuous connectivity for the perception component of the system.

However, local processing does not automatically guarantee a particular latency, cost, or accuracy. These properties depend on the selected camera, processor, AI model, software implementation, network architecture, and traffic-signal controller.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of four main components:

- 1) **Street Cameras:** Cameras capture images or video of traffic approaching an intersection.
- 2) **Smart Pole Computer:** A local computing device processes the camera data using a computer-vision model. It can estimate quantities such as vehicle presence, vehicle count, queue length, or traffic density.
- 3) **Traffic Signal Controller:** The local processing system provides traffic information or a control request to the signal controller. The controller is responsible for executing signal timing and safety logic.
- 4) **Cloud Server:** A cloud server can optionally be used for long-term data storage, monitoring, model training, or network-wide analysis. The cloud is not required to process every camera frame.

The conceptual data flow is therefore:

Street Cameras → *Smart Pole Computer* → *Traffic Signal Controller*

with an optional monitoring path:

Smart Pole Computer → *City Cloud Server*.

This architecture separates real-time intersection processing from optional cloud-based storage and analysis.

V. EMERGENCY VEHICLE SIGNAL PREEMPTION

An important correction to the original discussion concerns emergency vehicles. Emergency-vehicle signal preemption is not a new or generally unavailable capability. Traffic-signal guidance describes preemption as a special operating mode that can provide right-of-way to emergency vehicles. Detection can be implemented using technologies such as optical signals, radio communication, sound, pavement-based systems, or other methods [5].

Therefore, the research gap should not be stated as the absence of automatic emergency-vehicle path clearing. Instead, a more realistic research problem is the integration of computer-vision-based emergency-vehicle detection with existing signal-preemption mechanisms.

A possible local-AI system could detect an approaching emergency vehicle, estimate its direction of travel, and transmit a preemption request to the appropriate signal controller. For a route containing several intersections, the system could also investigate how preemption requests are coordinated between intersections. Such a system would need to address false detections, detection latency, communication failures, conflicting requests, pedestrian safety, and the requirements of the underlying traffic-signal controller.

It is also important to distinguish *signal preemption* from ordinary adaptive signal control. Adaptive control modifies signal timing in response to traffic conditions, whereas emergency-vehicle preemption temporarily transfers signal control to a special operating mode for the emergency vehicle [5].

VI. SYSTEM COMPARISON

Rather than assigning unsupported universal values for processing time and installation cost, the systems can be compared using qualitative and experimentally measurable characteristics.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Characteristic	Conventional	Local AI
Traffic input	Timing plans	Camera/sensor data
Adaptation	Limited	Possible in real time
Local processing	Usually low	Required
Network dependence	Application dependent	Can be reduced
AI capability	Not required	Possible
Emergency preemption	Possible	Can be integrated
Cloud processing	Not required	Optional

Actual latency, cost, accuracy, energy consumption, and network usage should be measured experimentally rather than assigned fixed values without a supporting study.

VII. RESEARCH GAP

Existing research demonstrates the use of adaptive traffic signal control, machine vision, and learning-based methods for traffic management [1], [2]. In addition, emergency-vehicle signal preemption is already an established traffic-management technique [5].

A more appropriate research gap is therefore the development and evaluation of a reliable local-AI system that combines these capabilities. In particular, future work can investigate:

- accurate emergency-vehicle detection using roadside cameras;
- low-latency local inference at intersections;
- safe communication between the AI system and the traffic-signal controller;
- coordination of emergency-vehicle requests across multiple intersections;
- handling of false positives and false negatives;
- comparison of local and cloud-based processing under realistic network conditions; and

- evaluation using measurable traffic and safety-related performance indicators.

These questions provide a more defensible research direction than assuming that existing traffic systems lack emergency-vehicle preemption.

VIII. CONCLUSION

Local AI provides a potential architecture for processing traffic-camera information close to the intersection. Existing research supports the use of adaptive signal control, machine vision, and learning-based methods for traffic management. However, claims about specific processing speeds, costs, or hardware capabilities must be supported by experimental measurements or reliable sources.

The proposed architecture places traffic perception and initial decision-making near the intersection while allowing optional cloud-based storage and analysis. Emergency-vehicle signal preemption should be treated as an existing traffic-management capability rather than as an entirely new function. The research opportunity is to investigate how local computer vision can be integrated safely and reliably with existing signal-control and preemption systems.

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